

Sylvain Carpentier

Seoul National University
sylcar@snu.ac.kr

Research Interests

Integrable systems, Lax operators, Poisson geometry, algebraic quantization, discrete integrable dynamics.

Employment

-BK21 Postdoctoral Fellow, Seoul National University	2024- present
-Senior Researcher, QSMS , Seoul National University	2022-2024
-Simons Junior Fellow, Columbia University	2017-2021

Education

-PhD in Mathematics, Massachusetts Institute of Technology , Advisor: Victor Kac	2013-2017
Thesis: <i>Rational Differential Operators and Integrable Systems of Partial Differential Equations</i>	
-Master of mathematics, Universite Pierre et Marie Curie	2011-2012
- Ecole Normale Supérieure de Paris	2009-2013

Awards and Fellowships

- Simons Junior Fellowship, Columbia University,	2017-2021
- Charles W. and Jennifer C. Johnson Prize, MIT	2017

Conference Organization:

- 2025 SNU workshop for representation theory and related topics, Incheon	08/2025
- 2024 SNU workshop for representation theory and related topics, Gangcheon	08/2024

Selected invited talks

-Korean Mathematics Society meeting	10/2025
-Ningbo, China, workshop on Poisson geometry and integrability,	09/2025
- Postech, Pohang, South Korea,	04/2024
- Universita della Sapienza, Roma,	02/2024
- <i>SusTech, Shenzhen, China, Efim Zelmanov's seminar</i>	12/2023
- <i>Conference on Algebraic Representation Theory</i> Taiwan NCTS,	12/2023
- <i>Incheon National University, 9th China Japan Korea International Conference</i> ,	08/2023
- Université de Dijon, France,	05/2023
- <i>Poisson and quantum structures in Integrable systems</i> at University of Leeds	05/2023
- Kolchin seminar, City University of New York	09/2022

Teaching Experience

- Seoul National University, graduate students mentoring,	2022-present
- Columbia University, Calculus II lecturer	2018
-MIT, teaching assistant for linear algebra	2016-2017
- MIT, Undergraduate Research mentor at SPUR program,	2015

Preprints

Arithmetic supports of Lax difference hierarchies, 2026
arXiv:2606.24273.

Algebraic quantisation approach to integrable differential-difference equations, 2025
with A. Mikhailov, J.P. Wang, arXiv 2509.21775.

Main Publications

Gelfand-Dickey realization of the supersymmetric W algebras for $gl(n+1,n)$ and $gl(n,n)$, 2025
with U. Suh, nonlinearity, 38, 11 .

Hamiltonians for the quantised Volterra hierarchy, 2024
with A. Mikhailov, J.P. Wang, nonlinearity 37 095033 .

Integrable systems on rectangular superalgebras via super Adler-type operators, 2024
with G. Lee, U. Suh, Communications in Mathematical Physics, Vol.405 No.7, p. 165.

Quantisations of the Volterra hierarchy, 2022
with A. Mikhailov, J.P. Wang, Lett Math Phys 112, 94.

Supersymmetric Bihamiltonian Integrable Systems, 2021
with U. Suh, Communications in Mathematical Physics, 382, 317-350.

p -reduced multicomponent KP hierarchy and classical W -algebras, 2020
with A. De Sole, V. Kac, D. Valeri and J. Van de Leur,
Communications in Mathematical Physics, 380 (2) 655-722.

PreHamiltonian and Hamiltonian operators for differential-difference equations, 2020
with A. Mikhailov and J.P. Wang, Nonlinearity, 33 (3) article Number 915.

Rational recursion operators for integrable differential-difference equations, 2019
with A. Mikhailov and J.P. Wang, Commun. Math. Phys. 370, 807–851.

Compatible Hamiltonian operators for the Krichever-Novikov equation, 2017
C. R. Math. Vol. 355, Issue 7, 744-747.

A sufficient condition for a Rational Differential Operator to generate an Integrable System, Japan. J. Math. 12, 33-89. 2017

Singular degree of a rational matrix pseudodifferential operator, 2015
with A. De Sole, V.G. Kac, Int. Math. Res. Not. IMRN 2015, no. 13, 5162 -5195.

Rational matrix pseudodifferential operators, 2013
with A. De Sole, V.G. Kac, Selecta Math. (N.S.) 20 (2014), no. 2, 403-419.

Some algebraic properties of differential operators 2012
with A. De Sole, V.G. Kac, J. Math. Phys. 53 (2012), no. 6, 063501, 12 pp.

Languages

French (native), English (fluent), Spanish (proficient), Korean (intermediate)